



Project Report

AI-Powered Campus Recruitment & Applicant Tracking System

Submitted By

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1. Executive Summary

This project presents an AI-Powered Campus Recruitment Portal designed to simplify the hiring process between students and recruiters. The system allows students to create profiles, submit resumes, apply for jobs, and receive job recommendations. Recruiters can post jobs, manage applications, and view ranked applicants.

The system is built using PostgreSQL, FastAPI, and HTML/CSS/JavaScript. Three AI agents were implemented for resume parsing, applicant ranking, and job recommendation. These agents automate the screening process and improve recruitment efficiency.

2. Introduction

Background

Campus recruitment is often manual and time-consuming. Recruiters review large numbers of resumes, while students struggle to find suitable opportunities. Existing job portals are not designed specifically for university hiring requirements such as CGPA filtering and skill-based matching.

This project was developed to create a dedicated recruitment platform for students and recruiters with AI-based decision-making features.

Objectives

- Extract skills from resumes automatically
- Rank applicants based on job requirements
- Recommend suitable jobs to students
- Maintain audit history of application updates
- Develop a complete web-based recruitment system

3. System Description

Main Features

- Student and recruiter role-based access
- Resume submission and skill extraction
- Job posting and application management
- AI-based applicant ranking
- Personalized job recommendations
- Audit logs for application updates

Innovations and Modifications

- Automatic resume parsing using keyword matching
- Weighted skill-based scoring system
- AI-generated top job recommendations
- Database triggers for enforcing business rules

4. AI Approach and Methodology

Agent 1 — Resume Keyword Matching Agent

This agent extracts skills and experience from resume text using regular expressions and keyword matching. Extracted skills are stored automatically in the database.

Agent 2 — Utility-Based Ranking Agent

This agent calculates a match score between students and job requirements using weighted skills and proficiency levels. Scores are generated automatically after every application.

Agent 3 — Goal-Based Recommendation Agent

This agent recommends the top matching jobs to students based on their profile, CGPA, and skills.

Heuristic Design

The scoring system uses:

- Skill importance weights
- Student proficiency levels
- Normalized score calculation between 0–100

Students with stronger matching skills receive higher scores.

5. System Mechanics and Rules

User Roles

Student

- Register and login
- Submit resume
- Apply for jobs
- View recommendations and applications

Recruiter

- Post and manage jobs
- View ranked applicants
- Update application status
- Access audit logs

Business Rules

- Applications to closed jobs are rejected
- Past deadlines are not allowed
- Duplicate applications are prevented
- Match scores are generated automatically through triggers

6. Implementation and Development

Development Process

The database schema and procedures were developed first, followed by backend APIs and frontend integration. Testing was performed through FastAPI Swagger UI.

Technologies Used

Backend

- Python
- FastAPI
- JWT Authentication

Database

- PostgreSQL
- psycopg2

Frontend

- HTML
- CSS
- JavaScript

Challenges Encountered

- Managing trigger execution order
- Handling resume parsing edge cases
- Enforcing CGPA validation at the database level

7. Team Contributions

Team Member	Responsibilities
Iqra Ashraf	Developed Agent 3 (Job Recommendation Agent) and recommendation APIs for generating personalized job suggestions based on student skills and CGPA. Also created Skills, Job_Posts, and Job_Required_Skills tables with seed data, developed Generate_Job_Recommendations and Update_Application_Status procedures, implemented Trigger 3 (AuditLog on Status Change) and Trigger 4 (Validate Job Deadline), created Top_Ranked_Applicants_View and Eligible_Students_View, and developed student job-related frontend pages and student.js.
Samiya Khan	Developed Agent 1 (Resume Keyword Matching Agent) for extracting skills and experience from resume text using keyword matching and regular expressions. Also handled project setup, authentication system, and student APIs, created Accounts, Companies, Students, and Recruiters tables with seed data, developed Apply_For_Job and Calculate_Match_Score procedures, implemented Trigger 1 (Auto Calculate Score) and Trigger 2 (Validate Job Active), created Active_Jobs_View, and developed frontend authentication pages along with student dashboard and profile pages.
Aseelah Masood	Created Student_Skills, Applications, Resumes, and AuditLog tables with Developed Agent 2 (Utility-Based Ranking Agent) for calculating weighted applicant match scores based on job requirements and skill proficiency. Also developed application-related APIs, created Student_Skills, Applications, Resumes, and AuditLog tables with seed data, implemented indexes,

	Application_Funnel_View, NOT NULL constraints, testing queries for triggers and procedures, and developed recruiter frontend pages including applicant management and audit log sections.
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8. Results and Discussion

The system was tested successfully through multiple API test cases.

- Agent 1 extracted skills and experience correctly from resumes
- Agent 2 generated accurate applicant match scores
- Agent 3 recommended relevant jobs based on student profiles
- Database triggers worked correctly for validation and audit logging

Limitations

- Resume parsing depends on exact keyword matches
- No PDF resume upload support
- Recommendations do not consider salary or location preferences

9. References

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